

AJAY VENKATESH

Brooklyn, NY



+1 516 585 9013



av3855@nyu.edu



linkedin.com/in/ajay-venkatesh-32013b17b



github.com/ajayv-19

Education

New York University | *M.S. Computer Engineering (GPA 3.9/4)*

Aug 2024 – Aug 2026

Visvesvaraya Technological University | *B.E. Electronics & Communication (GPA 8.9/10)*

Aug 2017 – Aug 2021

Experience

Amazon

May 2025 – Aug 2025

Software Development Engineer Intern

Seattle, US

- Designed enterprise-grade Pallet CV Console architecture with Midway SSO and HELIS authorization, enforcing role-based access control across Dev, Stakeholder, and FC Associate roles.
- Provisioned serverless backend using AWS CDK with API Gateway, Lambda (Java 21), DynamoDB, and S3; automated deployments through multi-stage CI/CD pipelines.
- Implemented React 18 frontend on Harmony with AWS UI components, role-specific views, and image filtering, delivering a faster and secure interface for 15K+ daily users.
- Built backend APIs with Java Coral, Smithy models, Dagger DI, and CBOR serialization; improved response latency by 35% and reliability through unit and integration testing.
- Validated cross-account security with CloudAuth tokens and IAM resource policies, reducing unauthorized access risks by 90% while meeting Amazon security standards.

New York University

Aug 2024 – May 2025

Software Developer (On-Campus)

New York, US

- Developed AI-powered Remote Patient Monitoring system with React Native, TypeScript, Cognito; project secured \$100K NSF grant for real-time health tracking.
- Built Python ETL pipelines on GCP Cloud Functions to convert NoSQL to SQL, applying parallel chunk processing to reduce transformation time by 60%.
- Created React web portal with QuickSight dashboards, Cognito authentication, and Stripe payments, enabling role-based data visualization for 50K+ health records.
- Engineered secure full-stack insurance broker platform with AWS Amplify, dynamic form generation, messaging features, and data dashboards for agent workflows.

PricewaterhouseCoopers (PwC)

Aug 2021 – Aug 2024

Senior Software Engineer

Bengaluru, IN

- Architected distributed backend on Microsoft Azure integrating Dynamics 365, achieving scalable, event-driven data processing with fault tolerance and high security.
- Mentored 5+ interns and led code reviews, reducing production bugs by 25% while accelerating Agile CI/CD delivery pipelines and improving team efficiency.
- Modernized legacy onboarding into modular B2B platform with data pipelines, event tracking, and BI reports, achieving 70% increase in automation and extensibility.
- Deployed OCR+NLP cognitive services via REST APIs for structured data extraction, reducing manual processing time by 75% and improving client integration.
- Implemented RBAC and optimized ETL pipelines processing 100K-row Excel files, reducing execution from 5 hours to 1 hour and enabling 80% efficiency gains.

Projects

Advanced Expense Tracker (JavaFX, JDBC, SQLite, JFreeChart)

- Designed modular OOP-based architecture (MVC, DAO) with secure login and group expense handling, ensuring clean separation of concerns and extensibility.
- Implemented concurrent transaction handling and real-time cost splitting with multithreading and synchronized blocks, adding analytics via JFreeChart.

Skills

Languages: Python, Java, C++, C, JavaScript, C#, SQL

Web Development: React, Node.js, Express, REST, GraphQL, Microservices

Databases: MySQL, PostgreSQL, MongoDB, JSON, Dataverse

Cloud: AWS (CDK, EC2, S3, DynamoDB, Lambda, API Gateway, Cognito, SageMaker), Azure (Logic Apps, Function Apps, AI Services, Data Lakes)

Tools: Git, Docker, Jupyter, Postman, Power BI, QuickSight, Power Automate, Visual Studio

Practices: Agile, Scrum, Waterfall, TDD, CI/CD

AI/ML: Neural Networks, PCA, LLMs, Classification, Pattern Recognition, MLOps

Certifications

AWS Cloud Practitioner

Microsoft Developer Associate

Power BI Data Analyst Associate

Microsoft Solution Architect Expert

Azure Data Scientist Associate

Azure Data Fundamentals

Publications

Supervised ML System Based Segmentation and Classification of Stroke Using Deep Learning Techniques (*IEEE*)

An Aquila-optimized SVM classifier for Diabetes prediction (*IEEE*)